

Adventist University of Central Africa

Master's in Big Data Analytics

Chapter 12 — Unsupervised Learning Lab

Introduction to Statistical Learning with Applications in R

Programme: MSc Big Data Analytics
Course Title: Data Mining
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1 Introduction

Supervised learning has a response variable that tells us what we are trying to predict. Unsupervised learning starts without that response. Instead, it looks for useful structure in the predictors themselves.

This lab develops the main tools from Chapter 12 of *An Introduction to Statistical Learning*:

1. principal component analysis (PCA), which replaces many correlated predictors with a smaller number of directions that explain their shared variation;
2. hierarchical clustering, which builds a tree of increasingly large clusters; and
3. K -means clustering, which directly searches for a chosen number of compact groups.

The methods answer related but different questions. PCA is primarily a continuous dimension-reduction method. Clustering is primarily a grouping method. Neither method automatically discovers the one true answer: the analyst must choose scales, distances, tuning values, and interpretations carefully.

2 Principal Component Analysis

2.1 Why dimension reduction is useful

Suppose a dataset contains p quantitative variables. Plotting or modelling all p variables can be difficult, especially when several variables measure similar underlying phenomena. PCA constructs new variables called principal components. Each component is a linear combination of the original variables:

$$Z_m = \phi_{1m}X_1 + \phi_{2m}X_2 + \cdots + \phi_{pm}X_p.$$

The loading vector $\phi_m = (\phi_{1m}, \dots, \phi_{pm})$ determines the direction of the component. PCA chooses the first direction to maximise the variance of the projected observations, subject to $\|\phi_1\| = 1$:

$$\phi_1 = \arg \max_{\|\phi\|=1} \text{Var}(X\phi).$$

The second component is the direction with the largest remaining variance, subject to being orthogonal to the first, and so on. The components are uncorrelated by construction and are ordered from greatest to least variance.

2.2 PCA with the USArrests data

USArrests contains four crime-related measurements for the 50 US states. The variables are measured on different scales, so the analysis should standardize them first. Otherwise, a variable with larger numerical units would influence the components more heavily even if it were not more informative.

```
data(USArrests)
summary(USArrests)
```

```
##      Murder      Assault      UrbanPop      Rape
## Min.      : 0.800   Min.      : 45.0   Min.      :32.00   Min.      : 7.30
## 1st Qu.: 4.075   1st Qu.:109.0   1st Qu.:54.50   1st Qu.:15.07
## Median : 7.250   Median :159.0   Median :66.00   Median :20.10
## Mean    : 7.788   Mean    :170.8   Mean     :65.54   Mean     :21.23
## 3rd Qu.:11.250   3rd Qu.:249.0   3rd Qu.:77.75   3rd Qu.:26.18
## Max.    :17.400   Max.    :337.0   Max.     :91.00   Max.     :46.00
```

```
USArrests_scaled <- scale(USArrests)
apply(USArrests_scaled, 2, mean)
```

```
##      Murder      Assault      UrbanPop      Rape
## 1.543210e-16 1.143530e-16 -3.996803e-16 8.526513e-16
```

```
apply(USArrests_scaled, 2, sd)
```

```
##      Murder      Assault      UrbanPop      Rape
##      1          1          1          1
```

The sample covariance matrix shows how the original variables move together. PCA diagonalizes this covariance matrix: its eigenvectors give the loading vectors, and its eigenvalues give the variances of the corresponding components.

```
usarrests.pca <- prcomp(USArrests, scale. = TRUE)
summary(usarrests.pca)
```

```
## Importance of components:
##
##              PC1      PC2      PC3      PC4
## Standard deviation    1.5749 0.9949 0.59713 0.41645
## Proportion of Variance 0.6201 0.2474 0.08914 0.04336
## Cumulative Proportion 0.6201 0.8675 0.95664 1.00000
```

The standard deviations reported by `prcomp()` are the square roots of the component variances. The proportion of variance explained by component m is therefore

$$\text{PVE}_m = \frac{d_m^2}{\sum_{j=1}^p d_j^2},$$

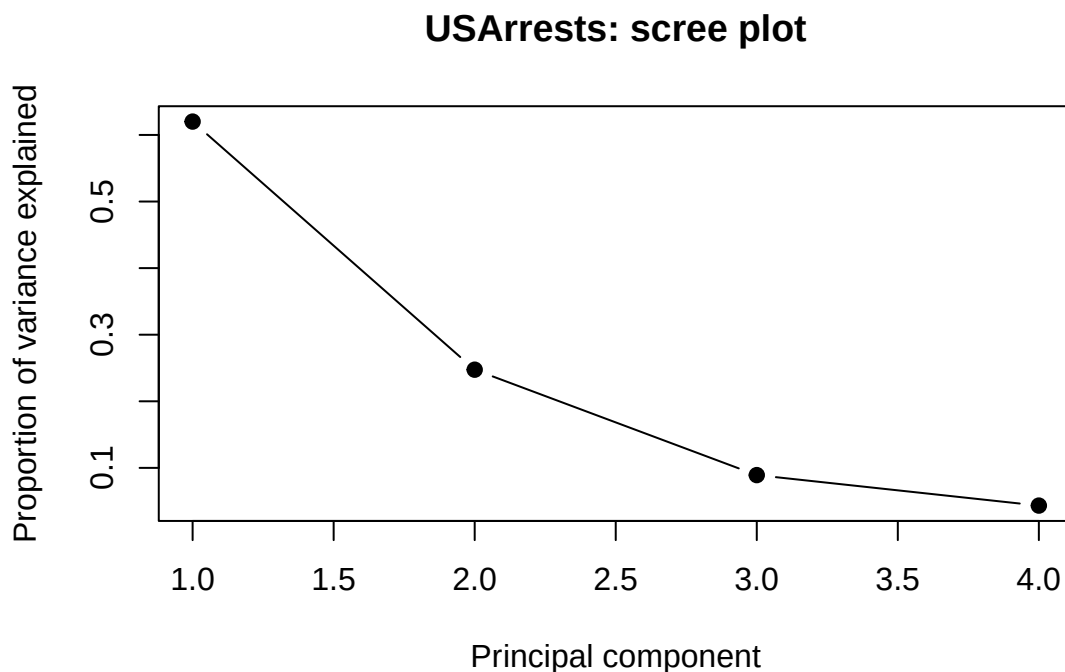
where d_m is the reported standard deviation of component m . The cumulative PVE tells us how many components are needed to retain a chosen amount of total variation.

```
usarrests_pve <- data.frame(
  Component = seq_along(usarrests.pca$sdev),
  Standard_deviation = usarrests.pca$sdev,
  Proportion_variance = usarrests.pca$sdev^2 /
    sum(usarrests.pca$sdev^2),
  Cumulative_variance = cumsum(usarrests.pca$sdev^2) /
    sum(usarrests.pca$sdev^2)
```

```
)
usarrests_pve

## Component Standard_deviation Proportion_variance Cumulative_variance
## 1 1 1.5748783 0.62006039 0.6200604
## 2 2 0.9948694 0.24744129 0.8675017
## 3 3 0.5971291 0.08914080 0.9566425
## 4 4 0.4164494 0.04335752 1.0000000

plot(usarrests_pve$Component, usarrests_pve$Proportion_variance,
     type = "b", pch = 19,
     xlab = "Principal component",
     ylab = "Proportion of variance explained",
     main = "USArrests: scree plot")
```



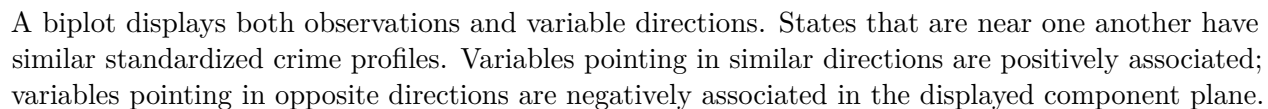
The loadings show which original variables contribute to each direction. A large absolute loading means that the variable is important for that component; the sign is a direction convention, because multiplying an entire component by -1 gives the same underlying axis.

```
round(usarrests.pca$rotation, 3)
```

```
## PC1 PC2 PC3 PC4
## Murder -0.536 -0.418 0.341 0.649
## Assault -0.583 -0.188 0.268 -0.743
## UrbanPop -0.278 0.873 0.378 0.134
## Rape -0.543 0.167 -0.818 0.089
```

The first component often places violent and property crime measures in a similar direction, while the second contrasts the variables that behave differently. This interpretation should be made from the loading pattern, not from the component name alone.

```
biplot(usarrests.pca, scale = 0, cex = 0.7,  
       main = "USArrests represented in the first two PCs")
```



3.1 Distances and linkage

$$d(i, j) = \sqrt{\sum_{k=1}^p (x_{ik} - x_{jk})^2}.$$

Hierarchical clustering initially treats every observation as its own cluster. At each step it merges the two closest clusters and records the fusion distance. The complete sequence of merges is represented by a dendrogram.

Different linkage rules define cluster distance differently. For clusters A and B :

$$D_{\text{single}}(A, B) = \min_{i \in A, j \in B} d(i, j),$$

$$D_{\text{complete}}(A, B) = \max_{i \in A, j \in B} d(i, j),$$

and average linkage uses the mean of all cross-cluster distances. Single linkage can join groups through chains of nearby observations, while complete linkage tends to favour compact groups.

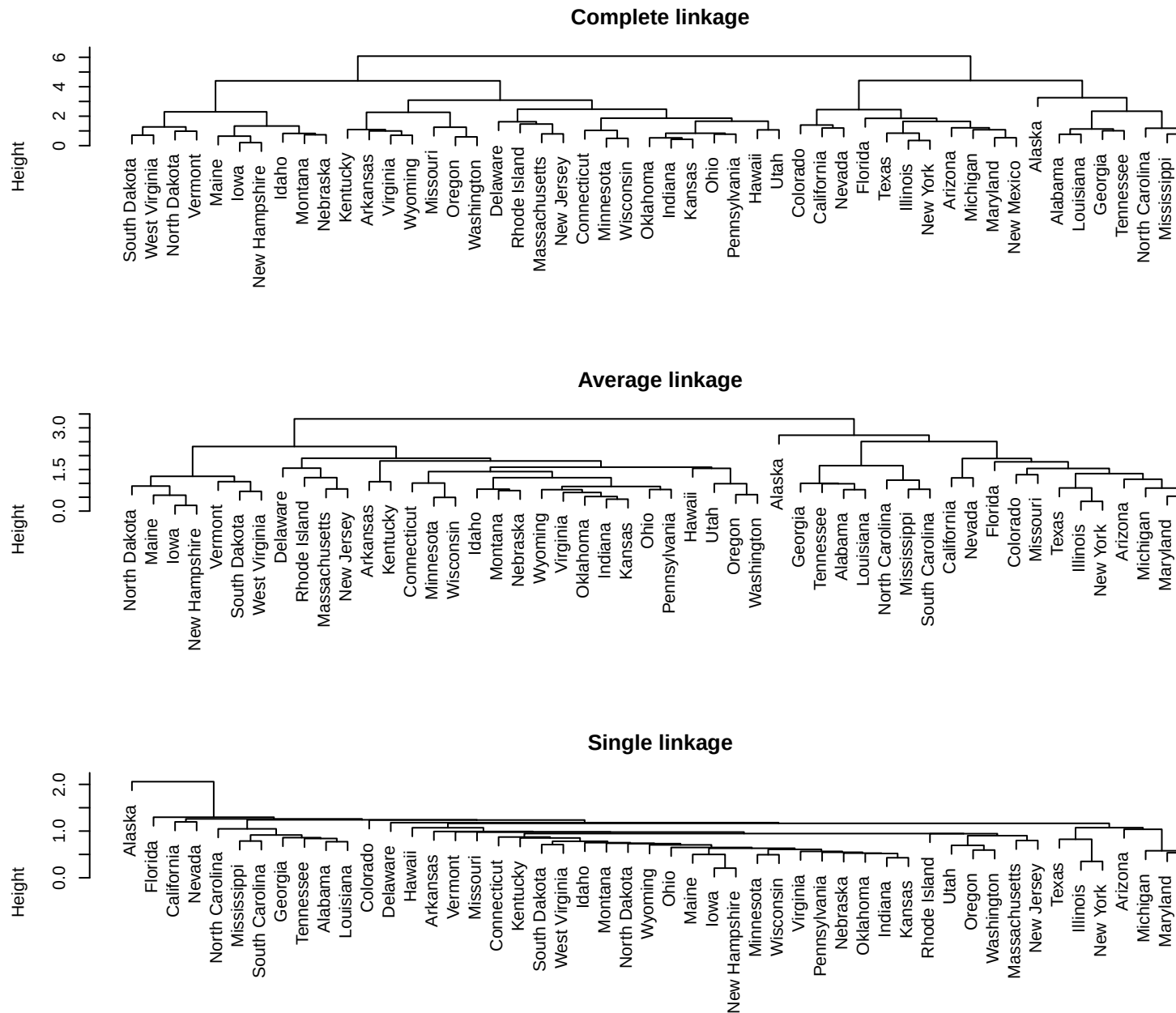
3.2 USArrests dendrograms

We use the standardized data so that the distance is not dominated by the measurement units of a single variable.

```
distance.usarrests <- dist(USArrests_scaled)

hc.complete <- hclust(distance.usarrests, method = "complete")
hc.average <- hclust(distance.usarrests, method = "average")
hc.single <- hclust(distance.usarrests, method = "single")

par(mfrow = c(3, 1), mar = c(3, 4, 3, 1))
plot(hc.complete, main = "Complete linkage", xlab = "", sub = "")
plot(hc.average, main = "Average linkage", xlab = "", sub = "")
plot(hc.single, main = "Single linkage", xlab = "", sub = "")
```



```
par(mfrow = c(1, 1))
```

The same observations can appear in visibly different positions in the three dendrograms because the linkage rule changes which merge is considered closest. A dendrogram is not a final partition until we choose a height or a number of clusters at which to cut it.

3.3 Cutting the dendrogram

```
complete_k4 <- cutree(hc.complete, k = 4)
average_k4 <- cutree(hc.average, k = 4)

head(complete_k4)
```

```
##      Alabama      Alaska      Arizona      Arkansas California      Colorado
##           1           1           2           3           2           2
```

```
table(complete_k4)
```

```
## complete_k4
##  1  2  3  4
##  8 11 21 10
```

```
table(average_k4)
```

```
## average_k4
##  1  2  3  4
##  7  1 12 30
```

```
comparison <- table(
  Complete_linkage = complete_k4,
  Average_linkage = average_k4
)
comparison
```

```
##              Average_linkage
## Complete_linkage  1  2  3  4
##              1  7  1  0  0
##              2  0  0 11  0
##              3  0  0  1 20
##              4  0  0  0 10
```

The cross-tabulation is a descriptive comparison, not a claim that one clustering is correct. If two methods place the same states together, that agreement suggests a stable grouping. Disagreement identifies observations whose group membership depends on the linkage choice.

3.4 A reproducible cluster interpretation

```
complete_profiles <- aggregate(
  USArrests,
  by = list(Cluster = complete_k4),
  FUN = mean
)
complete_profiles
```

```
##   Cluster  Murder  Assault UrbanPop  Rape
## 1       1 14.087500 252.7500 53.50000 24.53750
## 2       2 11.054545 264.0909 79.09091 32.61818
## 3       3  5.871429 134.4762 70.76190 18.58095
## 4       4  3.180000  78.7000 49.30000 11.63000
```

Cluster profiles translate an abstract dendrogram into the original units. A cluster with a high mean for **Murder** and **Assault**, for example, represents states with higher average values on those measures in this dataset. It is important to describe these as profiles rather than causal explanations.

4 *K*-Means Clustering

4.1 The objective

K-means requires us to choose the number of clusters K in advance. It then assigns each observation to one cluster and chooses cluster centres to minimise within-cluster sum of squares:

$$\text{WSS}(C_1, \dots, C_K) = \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \bar{x}_k\|^2.$$

The algorithm alternates between two steps:

1. assign each observation to its nearest current centre;
2. recompute each centre as the mean of the observations assigned to it.

Each step cannot increase the objective, but the procedure can stop at a local minimum. Consequently, `nstart` runs the algorithm from multiple random initialisations and keeps the solution with the smallest WSS.

4.2 Applying *K*-means to USArrests

```
set.seed(1)
kmeans_3 <- kmeans(USArrests_scaled, centers = 3, nstart = 20)
kmeans_3$size
```

```
## [1] 13 17 20
```

```
kmeans_3$centers
```

```
##      Murder    Assault  UrbanPop      Rape
## 1 -0.9615407 -1.1066010 -0.9301069 -0.9667633
## 2 -0.4469795 -0.3465138  0.4788049 -0.2571398
## 3  1.0049340  1.0138274  0.1975853  0.8469650
```

The centres are expressed in standardized units, so a centre of 1 means approximately one standard deviation above the overall feature mean. The cluster sizes show whether the solution is balanced or dominated by one large group.

```
kmeans_profiles <- aggregate(
  USArrests,
  by = list(Cluster = kmeans_3$cluster),
  FUN = mean
)
kmeans_profiles
```

```
##   Cluster    Murder    Assault  UrbanPop      Rape
## 1      1  3.600000  78.53846  52.07692 12.17692
## 2      2  5.841176 141.88235  72.47059 18.82353
## 3      3 12.165000 255.25000  68.40000 29.16500
```

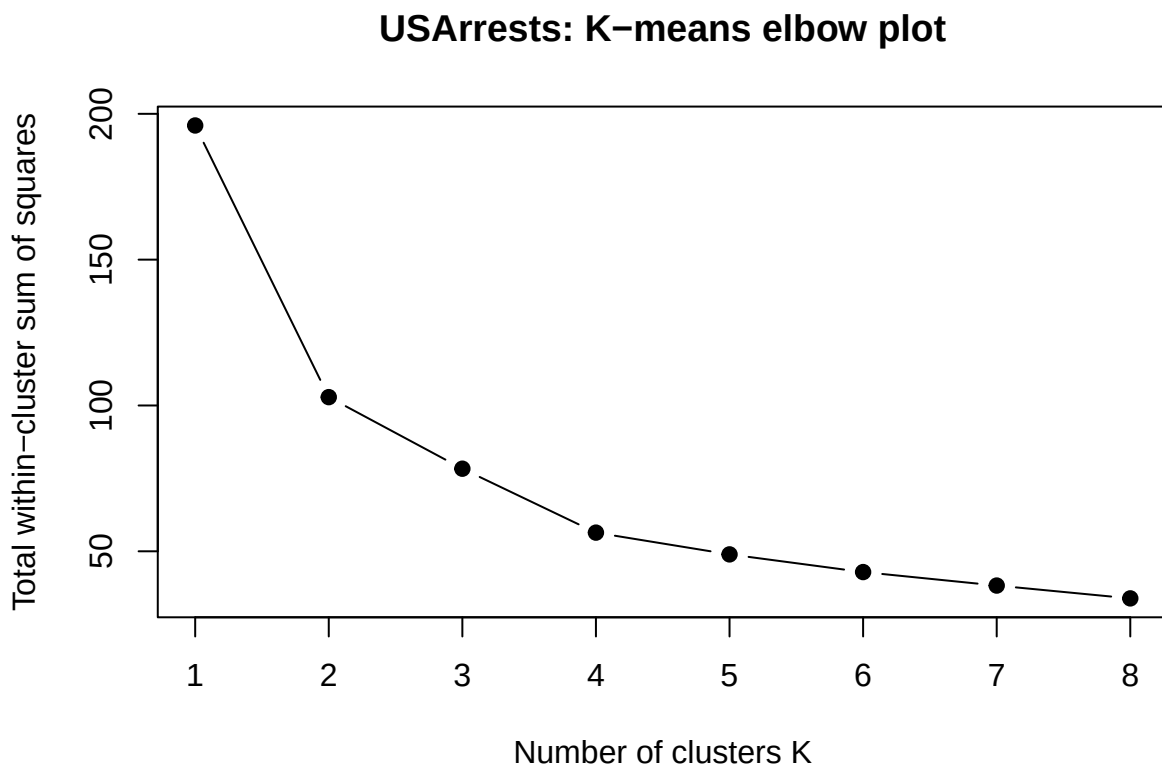
Unlike hierarchical clustering, K -means does not produce all possible group sizes in one fitted object. We must fit separate models for different values of K if we want to compare them.

4.3 Choosing K with an elbow plot

For each K , the total within-cluster sum of squares measures compactness. It will always decrease as K grows, because more centres give the algorithm more flexibility. We therefore look for an elbow: a point after which adding another cluster produces only modest improvement.

```
set.seed(1)
K_values <- 1:8
wss_values <- sapply(K_values, function(K) {
  kmeans(USArrests_scaled, centers = K, nstart = 20)$tot.withinss
})

plot(K_values, wss_values, type = "b", pch = 19,
     xlab = "Number of clusters K",
     ylab = "Total within-cluster sum of squares",
     main = "USArrests: K-means elbow plot")
```



The elbow is a judgement aid rather than a mechanical proof. In an unsupervised problem there is no response variable that can certify the best choice, so subject-matter usefulness and stability should also influence the choice.

5 The NCI60 Gene-Expression Data

5.1 Data structure

The NCI60 data contain gene-expression measurements for 64 cancer cell lines across thousands of genes. The data object has a numeric matrix and a label indicating the cancer type. We will use the labels for interpretation and checking, but unsupervised methods are fitted using the expression matrix.

```
data(NCI60)
str(NCI60)

## List of 2
## $ data: num [1:64, 1:6830] 0.3 0.68 0.94 0.28 0.485 ...
## ..- attr(*, "dimnames")=List of 2
## .. ..$ : chr [1:64] "V1" "V2" "V3" "V4" ...
## .. ..$ : chr [1:6830] "1" "2" "3" "4" ...
## $ labs: chr [1:64] "CNS" "CNS" "CNS" "RENAL" ...

dim(NCI60$data)

## [1] 64 6830

table(NCI60$labs)

##
##      BREAST      CNS      COLON K562A-repro K562B-repro      LEUKEMIA
##          7          5          7          1          1          6
## MCF7A-repro MCF7D-repro      MELANOMA      NSCLC      OVARIAN      PROSTATE
##          1          1          8          9          6          2
##      RENAL      UNKNOWN
##          9          1
```

The raw number of genes is much larger than the number of cell lines. PCA is therefore a natural first step: it gives a low-dimensional view of the variation before clustering.

5.2 PCA for NCI60

```
nci60.pca <- prcomp(NCI60$data, scale. = TRUE)
summary(nci60.pca)

## Importance of components:
##
##      PC1      PC2      PC3      PC4      PC5      PC6
## Standard deviation 27.8535 21.48136 19.82046 17.03256 15.97181 15.72108
## Proportion of Variance 0.1136 0.06756 0.05752 0.04248 0.03735 0.03619
## Cumulative Proportion 0.1136 0.18115 0.23867 0.28115 0.31850 0.35468
##
##      PC7      PC8      PC9      PC10      PC11      PC12
## Standard deviation 14.47145 13.54427 13.14400 12.73860 12.68672 12.15769
## Proportion of Variance 0.03066 0.02686 0.02529 0.02376 0.02357 0.02164
## Cumulative Proportion 0.38534 0.41220 0.43750 0.46126 0.48482 0.50646
##
##      PC13      PC14      PC15      PC16      PC17      PC18
```

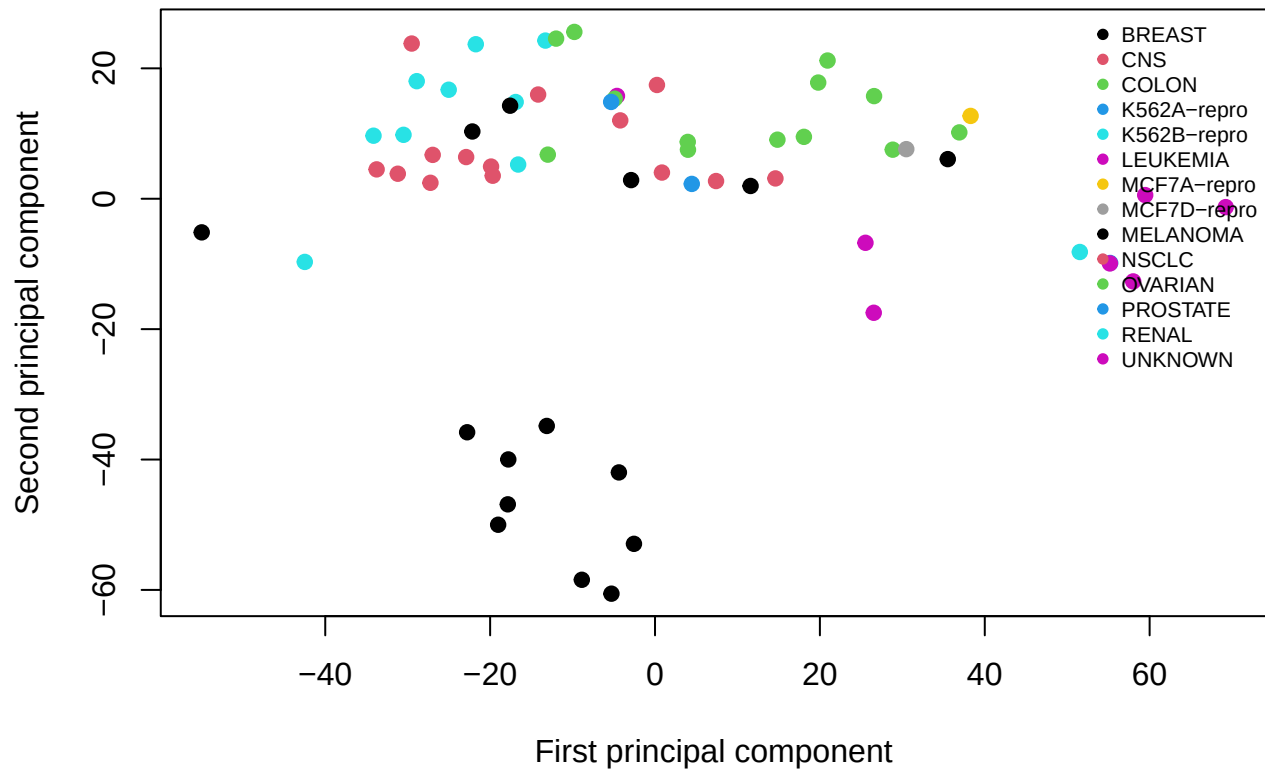
## Standard deviation	11.83019	11.62554	11.43779	11.00051	10.65666	10.48880	
## Proportion of Variance	0.02049	0.01979	0.01915	0.01772	0.01663	0.01611	
## Cumulative Proportion	0.52695	0.54674	0.56590	0.58361	0.60024	0.61635	
##	PC19	PC20	PC21	PC22	PC23	PC24	
## Standard deviation	10.43518	10.3219	10.14608	10.0544	9.90265	9.64766	
## Proportion of Variance	0.01594	0.0156	0.01507	0.0148	0.01436	0.01363	
## Cumulative Proportion	0.63229	0.6479	0.66296	0.6778	0.69212	0.70575	
##	PC25	PC26	PC27	PC28	PC29	PC30	PC31
## Standard deviation	9.50764	9.33253	9.27320	9.0900	8.98117	8.75003	8.59962
## Proportion of Variance	0.01324	0.01275	0.01259	0.0121	0.01181	0.01121	0.01083
## Cumulative Proportion	0.71899	0.73174	0.74433	0.7564	0.76824	0.77945	0.79027
##	PC32	PC33	PC34	PC35	PC36	PC37	PC38
## Standard deviation	8.44738	8.37305	8.21579	8.15731	7.97465	7.90446	7.82127
## Proportion of Variance	0.01045	0.01026	0.00988	0.00974	0.00931	0.00915	0.00896
## Cumulative Proportion	0.80072	0.81099	0.82087	0.83061	0.83992	0.84907	0.85803
##	PC39	PC40	PC41	PC42	PC43	PC44	PC45
## Standard deviation	7.72156	7.58603	7.45619	7.3444	7.10449	7.0131	6.95839
## Proportion of Variance	0.00873	0.00843	0.00814	0.0079	0.00739	0.0072	0.00709
## Cumulative Proportion	0.86676	0.87518	0.88332	0.8912	0.89861	0.9058	0.91290
##	PC46	PC47	PC48	PC49	PC50	PC51	PC52
## Standard deviation	6.8663	6.80744	6.64763	6.61607	6.40793	6.21984	6.20326
## Proportion of Variance	0.0069	0.00678	0.00647	0.00641	0.00601	0.00566	0.00563
## Cumulative Proportion	0.9198	0.92659	0.93306	0.93947	0.94548	0.95114	0.95678
##	PC53	PC54	PC55	PC56	PC57	PC58	PC59
## Standard deviation	6.06706	5.91805	5.91233	5.73539	5.47261	5.2921	5.02117
## Proportion of Variance	0.00539	0.00513	0.00512	0.00482	0.00438	0.0041	0.00369
## Cumulative Proportion	0.96216	0.96729	0.97241	0.97723	0.98161	0.9857	0.98940
##	PC60	PC61	PC62	PC63	PC64		
## Standard deviation	4.68398	4.17567	4.08212	4.04124	1.237e-14		
## Proportion of Variance	0.00321	0.00255	0.00244	0.00239	0.000e+00		
## Cumulative Proportion	0.99262	0.99517	0.99761	1.00000	1.000e+00		

```

nci60_scores <- nci60.pca$x[, 1:2]
nci60_labels <- sort(unique(as.character(NCI60$labs)))
nci60_label_codes <- match(as.character(NCI60$labs), nci60_labels)
plot(nci60_scores,
     col = nci60_label_codes, pch = 19,
     xlab = "First principal component",
     ylab = "Second principal component",
     main = "NCI60 cell lines in the first two PCs")
legend("topright", legend = nci60_labels,
     col = seq_along(nci60_labels), pch = 19,
     cex = 0.65, bty = "n")

```

NCI60 cell lines in the first two PCs



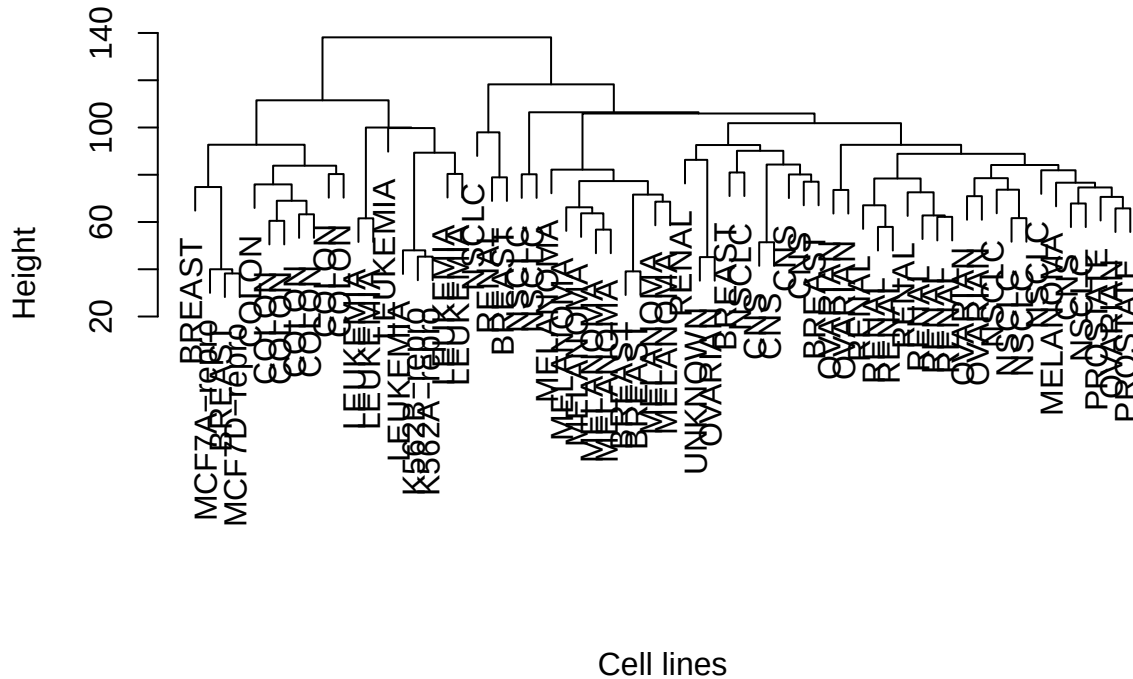
The colours are added after the unsupervised computation to see whether known cancer labels line up with visible structure. Strong overlap would mean that the first two PCs do not separate those types cleanly; visible grouping would suggest that the expression measurements contain type-related structure.

5.3 Hierarchical clustering for NCI60

```
nci60_distance <- dist(NCI60$data)
nci60_hc <- hclust(nci60_distance, method = "complete")

plot(nci60_hc, labels = NCI60$labs,
     main = "NCI60: complete-linkage clustering",
     xlab = "Cell lines", sub = "")
```

NCI60: complete-linkage clustering



```
nci60_clusters <- cutree(nci60_hc, k = 4)
table(Cluster = nci60_clusters, Cancer_type = NCI60$labs)
```

```
##      Cancer_type
## Cluster BREAST  CNS  COLON K562A-repro K562B-repro LEUKEMIA MCF7A-repro
##      1      4    5      0              0              0          0
##      2      1    0      0              0              0          0
##      3      0    0      0              1              1          6          0
##      4      2    0      7              0              0          0          1
##      Cancer_type
## Cluster MCF7D-repro MELANOMA NSCLC  OVARIAN  PROSTATE  RENAL  UNKNOWN
##      1              0          8      8          6          2      8          1
##      2              0          0      1          0          0      1          0
##      3              0          0      0          0          0      0          0
##      4              1          0      0          0          0      0          0
```

The table compares the unsupervised cluster assignments with the known labels. It is not a training confusion matrix: the labels were not used to create the clusters. Some clusters may contain several cancer types, and one cancer type may appear in multiple clusters, because the expression structure need not match the supplied labels exactly.

5.4 *K*-means for NCI60

```
set.seed(1)
nci60_kmeans <- kmeans(NCI60$data, centers = 4, nstart = 50)
table(Cluster = nci60_kmeans$cluster, Cancer_type = NCI60$labs)
```

```
##          Cancer_type
## Cluster BREAST CNS COLON K562A-repro K562B-repro LEUKEMIA MCF7A-repro
##      1      2  0    0          0          0      0          0
##      2      3  5    0          0          0      0          0
##      3      2  0    7          0          0      0          1
##      4      0  0    0          1          1      6          0
##          Cancer_type
## Cluster MCF7D-repro MELANOMA NSCLC OVARIAN PROSTATE RENAL UNKNOWN
##      1          0      7    0      0      0      0      0
##      2          0      1    6      4      1      9      1
##      3          1      0    3      2      1      0      0
##      4          0      0    0      0      0      0      0
```

The choice of $K = 4$ is made here for comparison with the hierarchical cut. In practice, we would examine several values of K , use stability or validation criteria, and consider the scientific purpose of the grouping.

6 Comparing the Methods

The methods can be summarised by the question each one answers:

Method	Main question	Main tuning choice
PCA	Which directions explain the most variation?	Number of components
Hierarchical clustering	How do observations merge at every distance level?	Distance, linkage, and cut
K -means	Which K compact groups minimise within-cluster variation?	K and initialisation

A responsible analysis reports the choices that produced the result. Scaling, distance, linkage, random seed, and the number of clusters are not cosmetic details; they define the geometry of the unsupervised problem.

7 Conclusion

PCA compresses correlated variables into orthogonal directions ordered by explained variance. Hierarchical clustering provides a nested picture of similarity, while K -means produces a direct partition for a selected K . The **USArrests** example shows how scaling and linkage affect a clustering result, and the **NCI60** example shows how PCA and clustering can reveal structure in high-dimensional gene-expression data.

Because unsupervised learning has no response variable to score directly, the best practice is to examine several reasonable specifications, report the sensitivity of the conclusions, and interpret the resulting patterns using subject-matter knowledge rather than treating one algorithmic output as an objective truth.